B. Deque Process

time limit per test 1.5 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: constructive algorithms, greedy, sortings, two pointers | https://codeforces.com/problemset/problem/2128/B

We say that an array a of size n is *bad* if and only if there exists $1 \le i \le n - 4$ such that one of the following conditions holds:

- a_i lt a_{i+1} lt a_{i+2} lt a_{i+3} lt a_{i+4}
- a_i gt a_{i+1} gt a_{i+2} gt a_{i+3} gt a_{i+4}

An array is *good* if and only if it's not *bad*. For example:

- a = [3, color*red*1, 2, 4, 5, 6] is *bad* because a_2 lt a_3 lt a_4 lt a_5 lt a_6.
- a = [color*red*7, 6, 5, 4, 1, 2, 3] is *bad* because a_1 gt a_2 gt a_3 gt a_4 gt a_5.
- a = [7, 6, 5, 1, 2, 3, 4] is *good*.

You're given a permutation[] $p_1, p_2, \ldots, p_n$.

You must perform n turns. At each turn, you must remove either the leftmost or the rightmost remaining element in p. Let q_i be the element removed at the i-th turn.

Choose which element to remove at each turn so that the resulting array q is *good*. We can show that under the given constraints, it's always possible.

[]A permutation of length n is an array consisting of n distinct integers from 1 to n in arbitrary order. For example, [2,3,1,5,4] is a permutation, but [1,2,2] is not a permutation (2 appears twice in the array), and [1,3,4] is also not a permutation (n=3 but there is 4 in the array).

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10\,000$). The description of the test cases follows.

The first line of each test case contains a single integer n ($5 \le n \le 100\,000$) — the length of the array.

The second line of each test case contains n integers $p_1, p_2, \ldots, p_n$ ($1 \le p_i \le n$, p_i are pairwise distinct) — elements of the permutation.

It is guaranteed that the sum of n over all test cases doesn't exceed 200 000.

Output

For each test case, you must output a string s of length n. For every $1 \le i \le n$, at the i-th turn:

- s_i = L means that you removed the leftmost element of p
- s_i = R means that you removed the rightmost element of p

We can show that an answer always exists. If there are multiple solutions, print any of them.

Example

Input

```
6
7
1 2 3 4 5 6 7
9
1 3 6 8 9 7 5 4 2
12
1 2 11 3 6 4 7 8 12 5 10 9
6
4 1 2 5 6 3
5
1 2 3 5 4
9
5 1 8 6 2 7 9 4 3
```

Output

```
RRRLLLL
LLRRLLRRL
LLLLLLLLLLL
LLLLLL
LLLLL
LLLLLLLLL
```

Note

In the first test case, the sequence color*blue*RRRcolor*red*LLLL results in q = [color*blue*7, color*blue*6, color*blue*5, color*red*1, color*red*2, color*red*3, color*red*4].

In the second test case, the sequence color*red*LLcolor*blue*RRcolor*red*LLcolor*blue*RRcolor*red*L results in q = [color*red*1, color*red*3, color*blue*2, color*blue*4, color*red*6, color*red*8, color*blue*5, color*blue*7, color*red*9].